T.I.P TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES
COMPUTER ENGINEERING
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COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	09/17/2026	**DATE SUBMITTED:**	09/17/2026
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ACTIVITY NO. 8.1 **Install, Configure, and Manage Enterprise Availability Monitoring via Ansible**

1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
   Role Structure:

```
erickalvarez@alvarezworkstation:~$ cd ~/Nagios_repo
erickalvarez@alvarezworkstation:~/Nagios_repo$ tree
.
├── ansible.cfg
├── centos_nagios.yml
├── installingnagios.yml
├── inventory.ini
├── nagios_site.yml
├── README.md
└── ubuntu_nagios.yml

0 directories, 7 files
erickalvarez@alvarezworkstation:~/Nagios_repo$
```

Observation: Here, I have divided dependencies (installingnagios.yml), nagios for Ubuntu, nagios for CentOS, and nagios_site to run ubuntu_nagios and centos_nagios simultaenously. I have divided it because my YML will become messy.

Inventory.ini:

```
erickalvarez@alvarezworkstation:~/Nagios_repo$ cat inventory.ini
[Ubuntu]
192.168.56.105 ansible_user=erickalvarez
192.168.56.107 ansible_user=erickalvarez

[CentOS]
192.168.56.108 ansible_user=ealvarez

[server1_lab]
192.168.56.120 ansible_user=alvarezworkstation

[server2_lab]
192.168.56.121 ansible_user=alvarezworkstation

[centos_lab]
192.168.56.122 ansible_user=ealvarez
erickalvarez@alvarezworkstation:~/Nagios_repo$
```

Observation: The inventory file has the IP address of the managed nodes running Ubuntu and CentOS. Ansible will utilize the inventory file to get information about the servers that need to install Nagios.

Nagios_site.yml:

```
erickalvarez@alvarezworkstation:~/Nagios_repo$ cat nagios_site.yml
---
- import_playbook: ubuntu_nagios.yml
- import_playbook: centos_nagios.yml
erickalvarez@alvarezworkstation:~/Nagios_repo$
```

Observation: The nagios_site.yml file is the primary playbook file. This file will call the Ubuntu and CentOS roles and make sure that Nagios is installed on both types of operating systems.

Dependencies:

- Ubuntu:

```
- name: Install Nagios dependencies on Ubuntu
  ansible.builtin.apt:
    name:
      - apache2
      - php
      - gcc
      - make
      - libc6
      - libc6-dev
      - libgd-dev
      - libssl-dev
      - libapache2-mod-php
      - unzip
      - wget
      - autoconf
      - openssl
      - apache2-utils
      - python3-passlib
    state: present
  when: ansible_distribution == "Ubuntu"
```

- CentOS:

```
- name: Install Nagios dependencies on CentOS
  ansible.builtin.dnf:
    name:
      - httpd
      - php
      - gcc
      - glibc
      - glibc-common
      - gd
      - gd-devel
      - make
      - openssl-devel
      - net-snmp
      - net-snmp-utils
      - unzip
      - wget
      - autoconf
      - httpd-tools
      - policycoreutils-python-utils
      - python3-pip
    state: present
  when: ansible_distribution == "CentOS"
```

```
- name: Install passlib via pip on CentOS
  ansible.builtin.pip:
    name: passlib
    executable: pip3
  when: ansible_distribution == "CentOS"
```

Observation: The dependency playbook installs the dependencies required by Nagios. Since the operating systems have different package managers, separate plays are included for both types of OSs.

Ansible Execution:

- Dependencies:

```
PLAY RECAP *********************************************************************
192.168.56.105             : ok=3    changed=2    unreachable=0    failed=0
   rescued=0    ignored=0
192.168.56.107             : ok=3    changed=2    unreachable=0    failed=0
   rescued=0    ignored=0
192.168.56.108             : ok=3    changed=2    unreachable=0    failed=0
   rescued=0    ignored=0
```

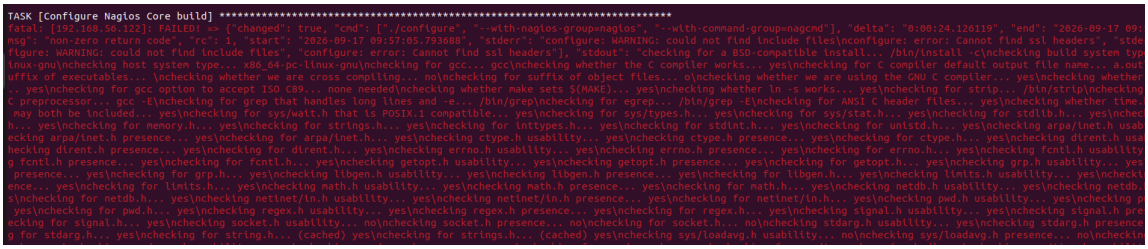
- Nagios installation:

```
changed: [192.168.56.108]

PLAY RECAP *********************************************************************
192.168.56.105             : ok=26   changed=5    unreachable=0    failed=0    skip
   rescued=0    ignored=0
192.168.56.107             : ok=26   changed=5    unreachable=0    failed=0    skip
   rescued=0    ignored=0
192.168.56.108             : ok=26   changed=8    unreachable=0    failed=0    skip
   rescued=0    ignored=0
```

Observation: The playbook was executed successfully in the managed nodes. After adding the missing dependency for SSL on CentOS, there were no further errors in the configuration of Nagios.

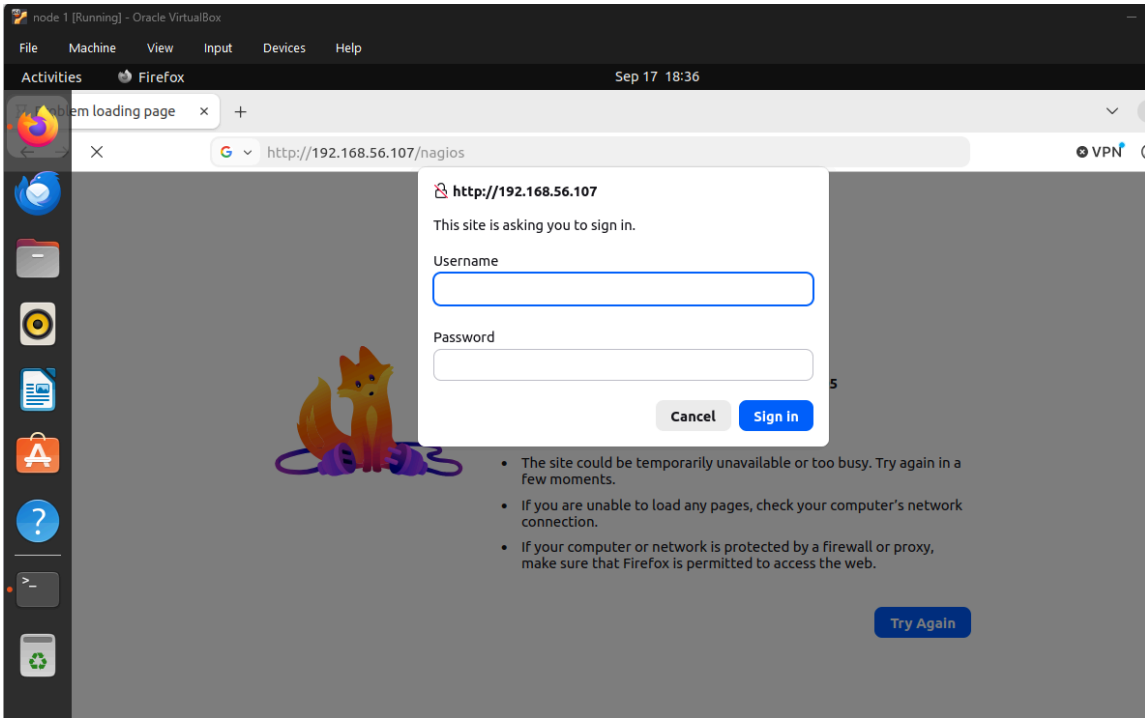
Error:



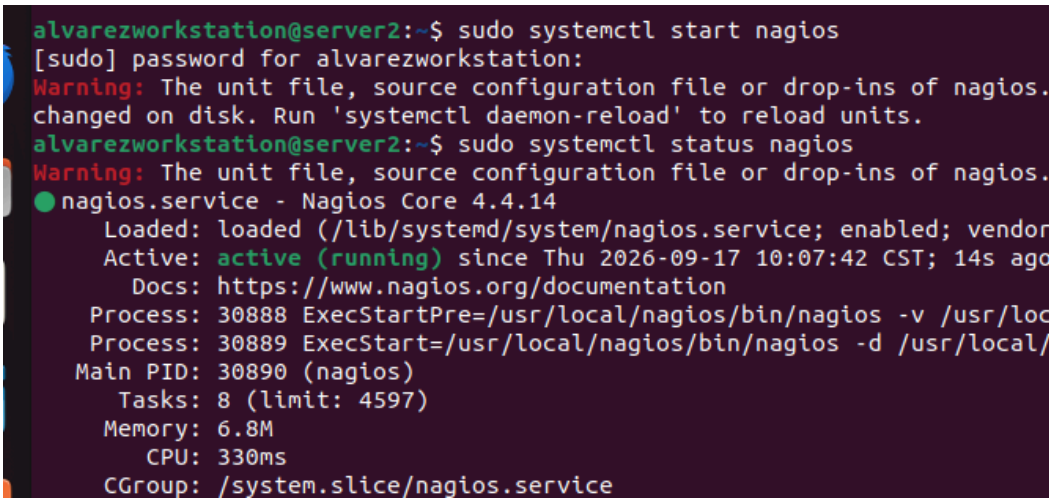
- While running the playbook, I experienced an issue while configuring Nagios on CentOS. Looking at the error message, I saw that the problem was related to SSL headers. When I checked the dependencies, I saw that openssl-devel was not present on the CentOS dependencies.

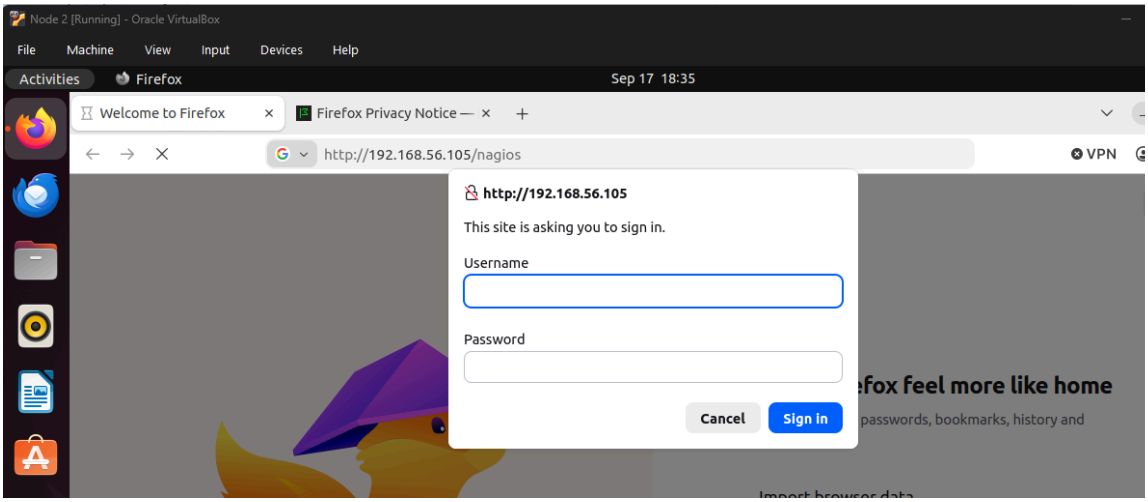
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
    - Ubuntu Server 1:

```
alvarezworkstation@ubuntu2:~$ sudo systemctl start nagios
alvarezworkstation@ubuntu2:~$ sudo systemctl status nagios
● nagios.service - Nagios Core 4.4.14
     Loaded: loaded (/lib/systemd/system/nagios.service; enabled; vendor pre
     Active: active (running) since Thu 2026-09-17 10:06:50 CST; 2s ago
       Docs: https://www.nagios.org/documentation
    Process: 30707 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/n
    Process: 30709 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagi
   Main PID: 30711 (nagios)
      Tasks: 10 (limit: 4597)
     Memory: 7.7M
        CPU: 343ms
     CGroup: /system.slice/nagios.service
             └─30711 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/n
```

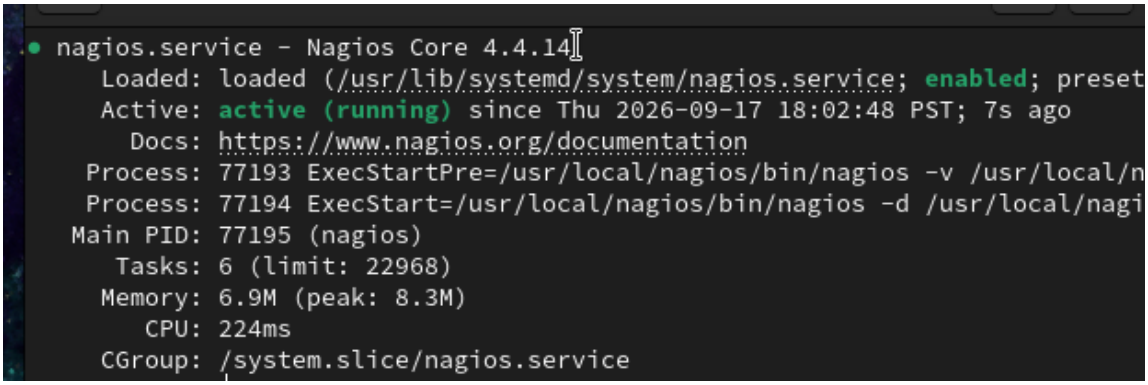


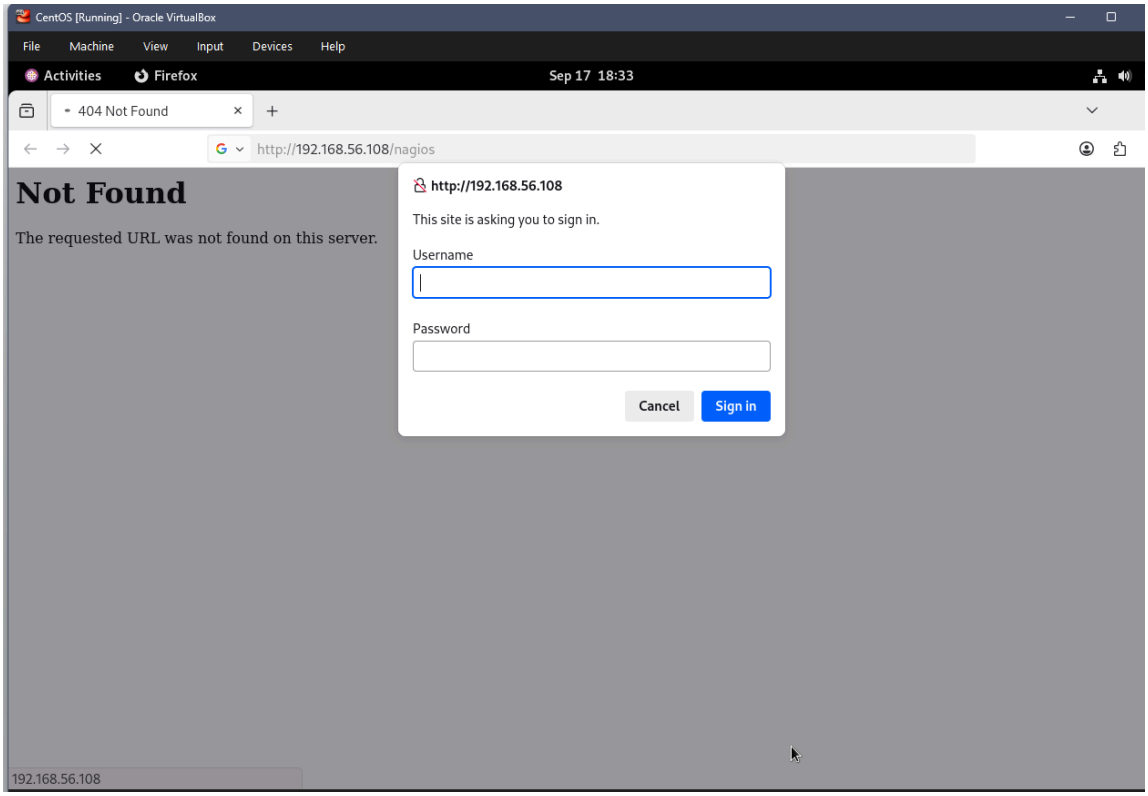
- Ubuntu Server 2:




Node 2 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Activities
Firefox
Sep 17 18:35
Welcome to Firefox
Firefox Privacy Notice —
http://192.168.56.105/nagios
VPN
http://192.168.56.105
This site is asking you to sign in.
Username
Password
Cancel
Sign in
fox feel more like home
passwords, bookmarks, history and

- CentOS:

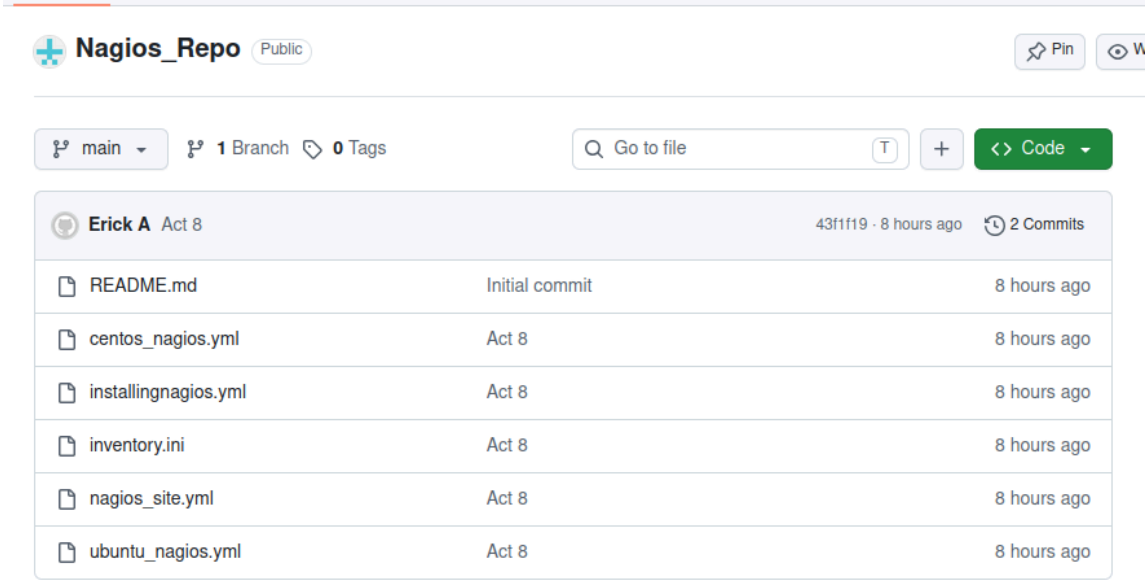

nagios.service - Nagios Core 4.4.14
Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset
Active: active (running) since Thu 2026-09-17 18:02:48 PST; 7s ago
Docs: https://www.nagios.org/documentation
Process: 77193 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/n
Process: 77194 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagi
Main PID: 77195 (nagios)
Tasks: 6 (limit: 22968)
Memory: 6.9M (peak: 8.3M)
CPU: 224ms
CGroup: /system.slice/nagios.service



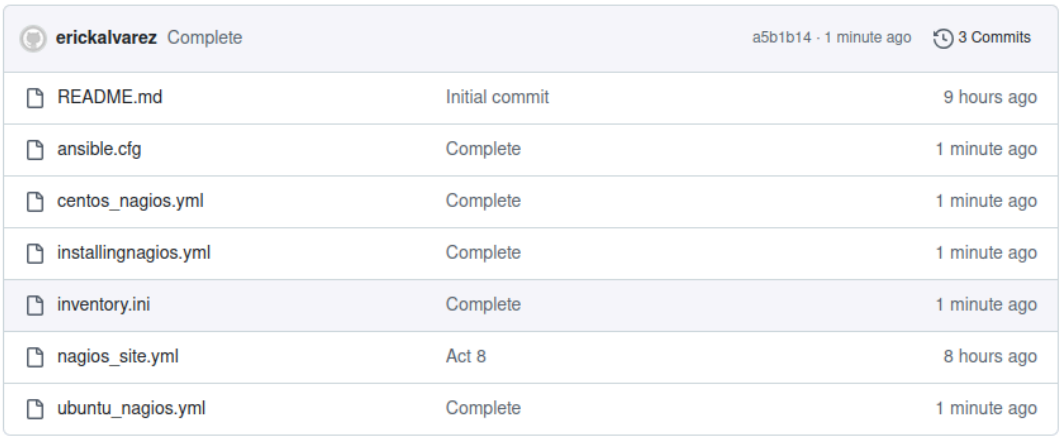
4. Make sure to create a new repository in GitHub for this activity.

```
erickalvarez@alvarezworkstation:~$ ls
CPE-212  Documents  Music         Pictures  snap       Videos
Desktop  Downloads  Nagios_repo   Public    Templates
erickalvarez@alvarezworkstation:~$
```

```
erickalvarez@alvarezworkstation:~$ cd Nagios_repo/
erickalvarez@alvarezworkstation:~/Nagios_repo$
```



After Git push:



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. **What are the benefits of having an availability monitoring tool?**
   - The purpose of availability monitoring software is to determine if a server, web page, or service is functioning correctly and is available for use. It can point out and report possible malfunctions or downtime, so that the responsible staff can solve problems before bigger problems arise. Moreover, availability monitoring helps us keep watch for the performance of the system as well as for the availability of key services.

Conclusions/Learnings:

This exercise helped me comprehend how monitoring availability can help keep servers connected. I also learned how Ansible can be utilized to help install Nagios across different OS systems including Ubuntu and CentOS. The

challenge I faced was in creating a playbook which turned out too long and complicated. To be more efficient, I decided to split all dependencies, Ubuntu, CentOS, and the main playbook into different files and roles. Although I still had many files and lines to deal with, it made my project more manageable.

While setting up Nagios, there were some problems that I faced, especially while accessing it through the browser. I got to know that there were certain dependencies and configurations needed, including some password module on CentOS. The toughest challenge was transferring all files to the new GitHub repository as it was not functioning initially. Hence I had to look up for solutions online and use AI to fix the issue. Overall, this exercise was pretty tough, since it made me learn many things about Ansible roles, debugging, and server monitoring, as well as project file management on GitHub.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT and Claude**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error explanation, troubleshooting, and grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was used only for error explanation, troubleshooting and grammar correction. The implementation and testing of the activity were completed by me.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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